Precept 6 Outline

ECO 202 TAs

October 15, 2026

1 Weekly Check In

1. Exam 2 grades released!
2. Any questions about the class, comments?
3. Reminders about resources:
   - Come to office hours for any preceptor, as much as you want.
     - Dexin: Tuesdays, 4–5pm, tables outside of JRR A19
     - Emily: Thursdays, 4:30–5:30pm, JRR A74
     - Ryan: Fridays, 2–3pm, JRR A74
   - Come to office hours of the professor: Fridays, 3pm–4pm, JRR 278
   - Check out the McGraw Center:
     - Individual tutoring: Link to page
     - Group study with a tutor: Link to page
   - Reach out to us by email if you've tried the above but are seeking support beyond these things.

2 Precept practice problems

1. **(Practice 10, Question 4)**

   For a stylized delivery business, let R be the next-day revenue change, measured in hundreds of dollars, with possible values 0 and 2. Let $Q = 1$ if all scheduled deliveries arrive on time and $Q = 0$ otherwise. Their joint distribution is

Delivery indicator	$R = 0$	$R = 2$	Marginal probability
$Q = 0$	0.30	0.10	
$Q = 1$	0.20	0.40	
Marginal probability			1.00

   (a) Complete all marginal probabilities and verify that the four joint cells form a valid distribution.

   (b) Write the marginal pmfs of R and Q, then calculate $\mathbb{E}[R]$ and $\mathbb{E}[Q]$.

   (c) Calculate $\text{Var}(R)$, $\text{SD}(R)$, $\text{Var}(Q)$, and $\text{SD}(Q)$.

   (d) Calculate $\mathbb{E}[RQ]$ directly from the joint table.

   (e) Calculate $\text{Cov}(R, Q)$ and $\text{Corr}(R, Q)$. State the units of each.

   (f) Test independence using the cell $(R = 2, Q = 1)$. Would the sign of the covariance alone be enough to establish independence or causality? Explain.

2. **(Practice 11, Question 2)**

   A two-record teaching population contains payment values 3 and 9, measured in tens of dollars. Select two records independently and uniformly with replacement and calculate $\overline{Y}_2$.

   (a) List all four equally likely ordered samples and calculate $\overline{Y}_2$ for each.

   (b) Construct the sampling distribution of $\overline{Y}_2$ and verify that its probabilities sum to one.

   (c) Calculate $\mathbb{E}[\overline{Y}_2]$, $\mathrm{Var}(\overline{Y}_2)$, and $\mathrm{SD}(\overline{Y}_2)$ directly from that distribution.

   (d) Calculate the population mean μ_Y and population variance σ_Y^2 of one uniform draw. Verify that $\mathbb{E}[\overline{Y}_2] = \mu_Y$ and $\mathrm{Var}(\overline{Y}_2) = \sigma_Y^2/2$.

   (e) If the realized ordered sample is $(3, 9)$, identify the realized sample distribution and realized sample mean. Explain which full sampling-distribution quantities do not change after observing it.

3. **(Practice 11, Question 3)**

   Under an explicitly stated independent with-replacement sampling model, let $H_i = 1$ when draw i has a specified characteristic and $H_i = 0$ otherwise, with $\mathbb{P}(H_i = 1) = p = 0.36$. Define

$$C_n = \sum_{i=1}^{n} H_i, \qquad \widehat{P}_n = \frac{C_n}{n}$$

   (a) Starting from the indicator mean and variance, derive $\mathbb{E}[C_n]$, $\mathrm{SD}(C_n)$, $\mathbb{E}[\widehat{P}_n]$, and $\mathrm{SE}(\widehat{P}_n)$.

   (b) Calculate all four quantities for $n = 25$.

   (c) Calculate all four quantities for $n = 100$.

   (d) Explain why every realized count is an integer although an expected count need not be.

   (e) When n is multiplied by four, state the exact factor change in the count center, count standard deviation, proportion center, and proportion standard error.

   (f) Identify which steps require independence and which expectation steps use linearity without requiring independence.

3 Extra practice problems

1. The random variables X and Y have marginal distributions

x	1	2
$\mathbb{P}(X = x)$	0.40	0.60

y	3	4
$\mathbb{P}(Y = y)$	0.30	0.70

   (a) Calculate $\mathrm{Var}(X)$.

   (b) Suppose X and Y are independent. Find the probability distribution of $W = Y - X$.

   (c) Suppose X and Y are not necessarily independent but retain the marginal distributions above. What is the largest possible value of $\mathbb{E}[W]$? Explain.

2. For working-age males in the United States, suppose the number of jobs held at the same time has distribution

Number of jobs	0	1	2
Probability	0.20	0.75	0.05

   For working-age females, suppose the distribution is

Number of jobs	0	1
Probability	0.50	0.50

Assume there are equally many working-age males and females.

(a) Randomly select one male and one female and calculate their average number of jobs. Find the probability distribution of this average.

(b) Randomly select one individual from all working-age people and ask for that person's number of jobs. Find its probability distribution.

3. Suppose X has distribution

x	-1	0	1
$\mathbb{P}(X = x)$	0.30	0.20	0.50

Let $X_1, X_2, \ldots$ be independent random variables with the same distribution as X, and define

$$\overline{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$$

(a) Compute $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

(b) What is the probability that $\overline{X}_2 > 0$?

(c) What values does $\overline{X}_{10}$ take with positive probability?

(d) Explain why $\sum_{i=1}^{n} X_i^2$ is exactly binomial, and give its parameters.

4. Let X and Y be random variables satisfying

$$\mathrm{Var}(X) = 3, \qquad \mathrm{Var}(Y) = 10, \qquad \mathrm{Var}(X + Y) = 9$$

What is the correlation between X and Y?

5. The random variable X has distribution

x	-2	4	5
$\mathbb{P}(X = x)$	0.30	0.50	0.20

(a) Compute the mean and variance of X.

(b) Let X_1 and X_2 be independent random variables with the same distribution as X. What is $\mathbb{P}(X_1 + X_2 < 0)$? Explain.

(c) A random variable Y satisfies $\mathbb{E}[X + Y] = 0$ and $\mathrm{Var}(X + Y) = 0$. What is the joint distribution of X and Y?

6. Suppose (Y, X) has joint distribution

	$X = -1$	$X = 1$	$X = 2$
$Y = 2$	0.10	0.35	0.20
$Y = 3$	0.05	0.20	0.10

(a) Are X and Y independent? Explain.

(b) What is $\mathrm{Var}(X)$? Explain.

(c) What is $\mathbb{P}(Y > X)$? Explain.

(d) Find the distribution of $W = X^2 - Y$.

7. A population consists of three individuals A, B, and C, whose values are -2, 0, and 4, respectively. A simple random sample of size two is taken without replacement. Find the sampling distribution of the sample mean and the median of that sampling distribution.

8. In a population of four individuals, a quantitative variable takes the values

$$\{1, 3, 4, 5\}$$

A simple random sample of size three is taken. Find the sampling distribution of the sample median.

9. Suppose x is a quantitative variable for a set of N individuals. Consider the random phenomenon of selecting one individual uniformly at random, and let X be the selected individual's value of x.

   (a) Describe the distribution of X in terms of the values $x_1, \ldots, x_N$.

   (b) What is $\mathbb{E}[X]$? Does it matter whether several individuals have the same value?

   (c) What is $\mathrm{Var}(X)$?

   (d) Now select n individuals independently with replacement. Let $X_1, \ldots, X_n$ be their values and let

$$\overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

   Find $\mathbb{E}[\overline{X}]$ and $\mathrm{Var}(\overline{X})$.